

Model-Based and Data-Driven Control of a Self-Balancing Robot Under Dynamic Conditions using Passivity

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Abstract—This project proposes a hybrid control workflow for a two-wheeled self-balancing robot. First, we use an existing robust model-based control system, which is simulated in MATLAB. Here we can test the model under various dynamic conditions, including collisions and weight additions. Once the model is tested under various input conditions, it will yield a large output sample. These data can be processed to create a data matrix for developing the data-driven control system. A data-driven control system is used in place of a Model-based approach as it is better adaptable to dynamic conditions. With known input-output data, the robot is able to better react to any input in real life. There are 2 types of data-driven control systems: Model-Based and Model-Free. We will conduct experiments on both systems to empirically evaluate which system demonstrates superior performance in power efficiency, computational speed, and space optimization. Passivity-based control employs convex energy shaping to achieve stability by enforcing passivity constraints and ensuring closed-loop energy dissipation during trajectory tracking. To enhance robustness, we will experiment with passivity-based control to achieve stability with minimal energy consumption.

Index Terms—Self-Balancing Robot, Model-Learning, Data Driven Control, Passivity, Dynamic Conditions.

I. INTRODUCTION

Self-balancing two-wheeled robots provide high maneuverability with zero turning radius and compact form, making them well-suited for confined environments. Their applications include warehouse automation, personal transport, surveillance, logistics, and indoor navigation in dynamic settings such as factories, mines, and crowded spaces.

However, the nonlinear, underactuated, and unstable dynamics of self-balancing robots make control challenging under disturbances such as terrain variations and payload changes. Conventional linear or PID-based methods often fail due to modeling inaccuracies and simulation-to-real-world gaps, motivating the need for robust and adaptive strategies.

This work proposes a hybrid control framework. First, a robust Active Disturbance Rejection Control (ADRC) baseline is developed and simulated in MATLAB. ADRC employs an Extended State Observer (ESO) to estimate and compensate for total disturbances, including uncertainties, friction, and external loads, without requiring precise models. For two-wheeled robots, nonlinear ADRC decouples angle and velocity

loops, enabling fast disturbance rejection (settling < 1 s) and robustness to $\pm 20\%$ parameter variations, outperforming classical PID and pole-placement methods [1].

The generated data are then utilized for model-free data-driven control, such as intelligent PD (iPD) and iPD-SMC. These approaches rely solely on input-output data to regulate tracking error, handling nonlinearities and disturbances more effectively than traditional LQR or PID methods while maintaining fast settling and low overshoot [2].

Finally, passivity-based control is enforced via frequency-domain convex optimization to ensure stability and safe operation. Using measured frequency-response data, a fixed-structure controller is synthesized under positive-real constraints, guaranteeing energy dissipation and robustness without explicit system models [5].

This pipeline builds progressively: ADRC stabilizes, MFC provides baseline control, and passivity certifies safety. Yields robust balancing with $< 3\%$ error under disturbances.

II. PROBLEM DESCRIPTION

Two-wheeled self-balancing robots are nonlinear, underactuated, and inherently unstable systems, modeled as wheeled inverted pendulums as stated in [1] [7]. The dynamics are given by

$$(M + m)\ddot{x} + b\dot{x} + m\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = F \quad (1)$$

$$(I + ml^2)\ddot{\theta} + mgl\sin\theta = -ml\ddot{x}\cos\theta \quad (2)$$

where M is cart mass, m body mass, l pendulum length, x position, θ tilt, F wheel force. Linearization around $\theta = 0$ yields unstable poles, which are sensitive to parameters [1] [7].

To facilitate control design, the nonlinear dynamics in (1) and (2) are embedded within a structured control architecture as shown in Fig. 1 [1]. These equations define the nonlinear plant, where the wheel-generated force F acts as the input and the states $(x, \dot{x}, \theta, \dot{\theta})$ evolve according to coupled cart-pendulum dynamics.

In Fig. 1, the plant is shown as a subsystem, while the remaining blocks represent sensing, control, and actuation.

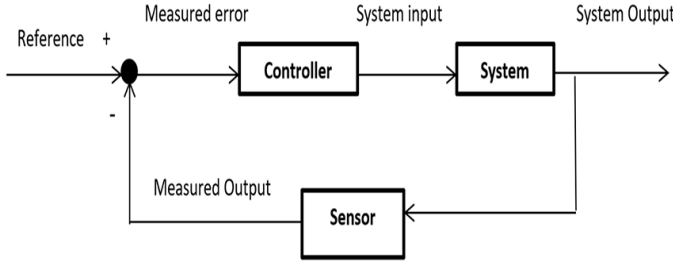


Fig. 1. Control System Block Diagram

IMU measurements of θ and $\dot{\theta}$ are fed to the controller, which computes the control input. This input is then distributed to the wheels via allocation and motor control, closing the feedback loop.

This representation abstracts the nonlinear dynamics while preserving their influence on the closed-loop behavior.

Key challenges include:

- **Instability:** Fast unstable eigenvalues demand high-bandwidth control.
- **Disturbances:** Pushes/terrain cause drift; linear controllers fail beyond small angles.
- **Uncertainties:** Friction, payload shifts alter CoM/inertia; neglected couplings degrade performance.
- **Nonlinearities:** Large tilts require global control; MPC computationally heavy.
- **Drift/Positioning:** Balance induces velocity bias without secondary loops.

A. Model-Based Limitations

Model-based control relies on accurate system models, which are often difficult to obtain for complex systems due to nonlinearities, uncertainties, and time-varying dynamics. Consequently, modeling errors can degrade performance and compromise stability [2] [3]. Most approaches rely on linearization around equilibrium points, limiting their validity to small deviations and leading to degraded performance under large disturbances or varying conditions [2] [1]. Although robust methods attempt to address uncertainties, they introduce conservatism due to difficulty in accurately bounding them, resulting in suboptimal performance [5].

B. Data-Driven Control Limitations

Data-driven control methods have several limitations. They require sufficiently informative data, typically ensured through persistent excitation; inadequate excitation can lead to infeasible designs [6]. In practice, excitation is constrained by actuator limits and safety requirements, making performance highly dependent on data quality, especially in unstable systems [2] [3]. These methods are also sensitive to measurement noise, as controller synthesis relies directly on data, while robust formulations depend on assumptions about noise bounds that are difficult to estimate [6] [1]. Moreover, many approaches rely on linear assumptions, reducing effectiveness under large disturbances or operation outside the data region

[2] [1]. Finally, data-driven control often involves large-scale optimization and parameter tuning, increasing computational burden and limiting scalability and real-time applicability [5].

C. Hybrid Solution Motivation

This approach fixes both:

- **ADRC:** Robust baseline + data-accurate dynamics
- **MFC:** Model-free performance
- **Passivity SDP:** Mathematical safety certification

Therefore, when we look into the flowchart, Fig. 2.

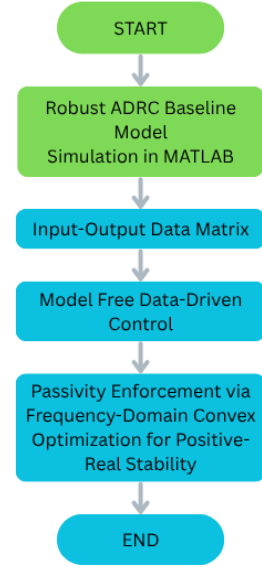


Fig. 2. Project Flowchart

III. MATHEMATICAL MODEL

The mathematical model of the two-wheeled self-balancing robot is based on the wheeled inverted pendulum configuration, with Fig. 3 as the reference. The system dynamics are derived using Newton's Second Law, $\sum F = ma$ and $\sum M = I\alpha$. The control objective is to maintain the upright equilibrium, where all accelerations are zero, i.e., $\sum F = 0$ and $\sum M = 0$.

Unlike the classical cart-pendulum system, the robot employs two independently actuated motors in a differential-drive configuration. This enables both forward motion and yaw control, introducing additional coupling between translational, rotational, and steering dynamics.

Applying Newton's laws to the system in Fig. 3, the force and moment balances are obtained. The horizontal force balance is given by

$$\sum f_x = 0 \iff F - b\dot{x} - (M + m)\ddot{x} - ml\ddot{\theta} \cos \theta + ml\dot{\theta}^2 \sin \theta = 0 \quad (3)$$

which captures the driving force, viscous friction, inertial effects, and coupling between translational and rotational motion.

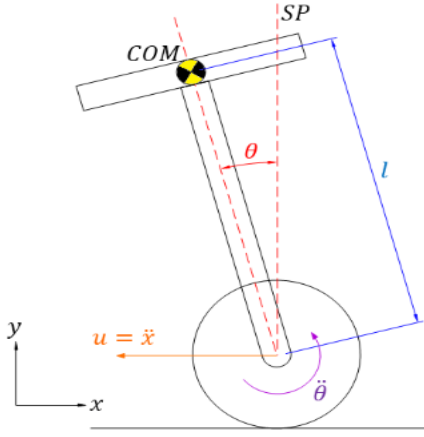


Fig. 3. Dynamics of Self-Balancing Robot

The vertical force balance is expressed as

$$\sum f_y = 0 \iff Mg + mg - ml\ddot{\theta} \sin \theta - ml\dot{\theta}^2 \cos \theta = 0 \quad (4)$$

which enforces the no-lift-off constraint through gravitational and inertial contributions.

The pitching dynamics about the wheel axle are obtained from the moment balance

$$\sum M_A = 0 \iff mgl \sin \theta + I\ddot{\theta} + ml\ddot{x} \cos \theta = 0 \quad (5)$$

where the gravitational torque, rotational inertia, and coupling with chassis acceleration determine the balancing behavior.

Additionally, the yaw dynamics arise from differential wheel forces and are given by

$$\sum M_z = 0 \iff l_w(F_R - F_L) - I_z\ddot{\psi} = 0 \quad (6)$$

which enables steering through the difference in left and right wheel forces.

Together, (3)–(6) describe the full nonlinear dynamics of the system. In this work, emphasis is placed on the longitudinal and pitching dynamics (3) and (5), as they directly govern balance, while yaw dynamics are acknowledged but not explicitly considered to reduce model complexity.

A. Nonlinear State-Space Model

The nonlinear dynamics of the system are expressed in a control-affine state-space form as

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})u, \quad (7)$$

where u represents the wheel driving force F . The state vector is defined as

$$\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}. \quad (8)$$

From this definition, two of the state equations follow directly:

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = \ddot{x}, \quad \dot{x}_3 = x_4, \quad \dot{x}_4 = \ddot{\theta}. \quad (9)$$

Rearranging the (3) and (5), the coupled dynamics can be written as

$$(M + m)\ddot{x} - ml\ddot{\theta} \cos \theta = r_1(x, u), \quad (10)$$

$$(I + ml^2)\ddot{\theta} - ml\ddot{x} \cos \theta = r_2(x), \quad (11)$$

where

$$r_1(x, u) = u - bx_2 - mlx_4^2 \sin x_3, \quad r_2(x) = -mgl \sin x_3. \quad (12)$$

These equations represent a coupled second-order system, where translational and rotational accelerations are interdependent. To explicitly solve for $\ddot{x}$ and $\ddot{\theta}$, the system is written in matrix form as

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}, \quad (13)$$

with

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} M + m & -ml \cos \theta \\ ml \cos \theta & I + ml^2 \end{bmatrix}. \quad (14)$$

By inverting this matrix, the accelerations are obtained as

$$\ddot{x} = \frac{a_{11}r_1 + a_{12}r_2}{\Delta}, \quad \ddot{\theta} = \frac{a_{21}r_1 + a_{22}r_2}{\Delta}, \quad (15)$$

where $\Delta = a_{11}a_{22} - a_{12}a_{21}$. These expressions explicitly show how the nonlinear forces, gravity, and control input influence the system accelerations.

Substituting these results into the state equations yields the nonlinear state-space model

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} x_2 \\ f_1(x) \\ x_4 \\ f_2(x) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} 0 \\ g_1(x) \\ 0 \\ g_2(x) \end{bmatrix}}_{g(x)} u, \quad (16)$$

where $f(x)$ represents the natural nonlinear dynamics, including gravitational and coupling effects, and $g(x)$ describes how the input force influences the system through the mass-inertia coupling.

This control-affine formulation separates the intrinsic system behavior from the control input, making it suitable for nonlinear control design. With the nonlinear state-space model established, the next step is to linearize the system about the upright equilibrium to obtain a tractable model for controller synthesis.

B. Linearization of Nonlinear Equations

To enable linear control design, the nonlinear model is linearized about the upright equilibrium to obtain

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu. \quad (17)$$

The angular coordinate is redefined as $\theta = \pi + \phi$, where $\phi \approx 0$. Using small-angle approximations,

$$\sin(\pi + \phi) \approx -\phi, \quad \cos(\pi + \phi) \approx -1, \quad (18)$$

the nonlinear dynamics are simplified near the equilibrium. Substituting and neglecting higher-order terms yields

$$(M + m)\ddot{x} + m\ddot{\phi} + b\dot{x} = u, \quad (I + ml^2)\ddot{\phi} - mgl\phi = ml\ddot{x}. \quad (19)$$

Define $a = M + m$, $c = ml$, $d = I + ml^2$, to rewrite

$$a\ddot{x} + c\ddot{\phi} = u - b\dot{x}, \quad -c\ddot{x} + d\ddot{\phi} = mgl\phi. \quad (20)$$

Solving gives

$$\ddot{x} = \frac{d(u - b\dot{x}) - cmgl\phi}{\Delta}, \quad \ddot{\phi} = \frac{c(u - b\dot{x}) + amgl\phi}{\Delta}, \quad (21)$$

where $\Delta = ad + c^2$. With state vector (8) the linearized state-space model is

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{db}{\Delta} & -\frac{cmgl}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{cb}{\Delta} & \frac{amgl}{\Delta} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{d}{\Delta} \\ 0 \\ \frac{c}{\Delta} \end{bmatrix}. \quad (22)$$

The matrix A captures the coupled dynamics and gravitational instability, while B describes the input influence. This model forms the basis for subsequent state-feedback and LMI-based control design.

C. State-Feedback Stabilization

To stabilize the linearized system, consider the LTI model

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + Bu(t). \quad (23)$$

The objective is to design $u(t)$ such that the states converge to equilibrium. A full-state feedback law is defined as

$$u(t) = K\mathbf{x}(t), \quad (24)$$

where K is the gain matrix. Substitution yields the closed-loop system

$$\dot{\mathbf{x}}(t) = (A + BK)\mathbf{x}(t), \quad (25)$$

so stability depends on the eigenvalues of $A + BK$.

Using Lyapunov theory, consider $V(x) = x^T Px$, with $P > 0$. Asymptotic stability is guaranteed if

$$(A + BK)^T P + P(A + BK) < 0. \quad (26)$$

This ensures energy decay and convergence to equilibrium. However, the condition is nonlinear in K and P , motivating its reformulation as an LMI for convex optimization.

D. Lyapunov/LMI-Based Control Design

The Lyapunov stability condition ensures asymptotic stability of the closed-loop system. However, it is bilinear in P and K , making it non-convex.

To obtain a convex formulation, define

$$Y = KP \iff K = YP^{-1} \quad (27)$$

Substituting (27) into (26) and multiplying by P gives

$$(AP + BY) + (AP + BY)^T < 0, \quad P > 0. \quad (28)$$

This condition is linear in P and Y , forming an LMI. The LMI can be solved via semidefinite programming, and the feedback gain is recovered as (27).

For the two-wheeled self-balancing robot, substituting the linearized matrices A and B yields a stabilizing controller. This approach is classified as model-based LMI control.

E. Data-Driven LMI Control Design

Model-based LMI design requires knowledge of (A, B) , which may be unavailable due to uncertainties. To address this, stabilization is formulated directly using measured input-state data.

Assume collected trajectory data satisfy

$$X_1 = AX_0 + BU_0, \quad (29)$$

where X_0 , X_1 , and U_0 denote state and input data. This relation implicitly represents the system dynamics without explicit identification.

Substituting this into the Lyapunov framework yields the data-driven LMI

$$\begin{bmatrix} X_0 Q & X_1 Q \\ Q^T X_1^T & X_0 Q \end{bmatrix} > 0, \quad (30)$$

where Q is the decision variable.

The LMI is convex and solved via semidefinite programming. The stabilizing gain is then recovered as

$$K = U_0 Q (X_0 Q)^{-1}. \quad (31)$$

Thus, Lyapunov stability is preserved while replacing model-based dynamics with data, enabling stabilization without explicit system identification.

IV. SIMULATIONS

To evaluate controller performance, simulations were conducted in MATLAB using both the linearized and nonlinear models of the system. The objective is to validate the LMI-based controller and compare the behavior of the linear approximation with the full nonlinear dynamics.

1) *Simulation Setup*: The controller gain K was obtained by solving the Lyapunov-based LMI stability condition, and the control law in (24) was used throughout, where the state vector is (8).

Simulations were initialized with a large disturbance: $\mathbf{x}(0) = [0 \ 0 \ 45^\circ \ 0]^T$, representing a significant angular deviation with zero initial position and velocities.

To assess robustness, the friction coefficient b and inertia I were randomly varied within predefined ranges across simulations, emulating practical parameter uncertainties.

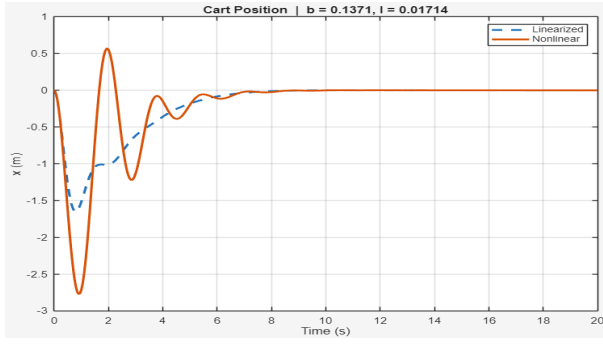


Fig. 4. Cart position response for 45° initial angle.

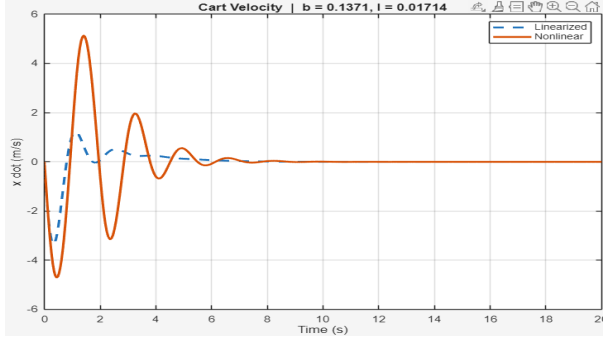


Fig. 5. Cart velocity response for 45° initial angle.

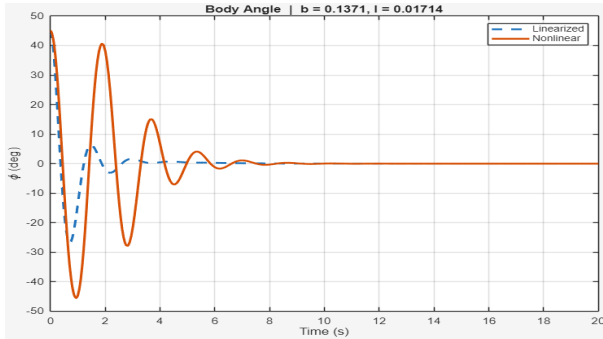


Fig. 6. Body angle response for 45° initial angle.

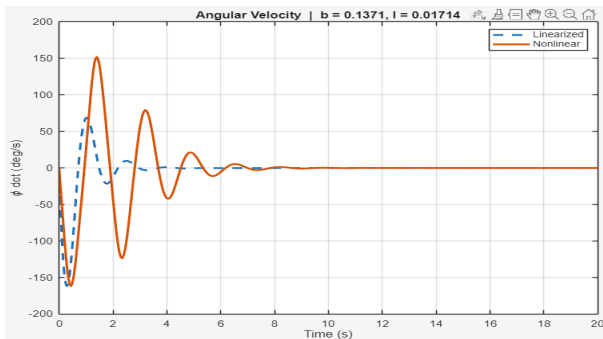


Fig. 7. Angular velocity response for 45° initial angle.

sponses of the linearized and nonlinear models were compared for $x(t)$, $\dot{x}(t)$, $\phi(t)$, and $\dot{\phi}(t)$.

For larger initial conditions (e.g., 45°), noticeable deviations arise during transients, reflecting nonlinear effects. Nevertheless, the controller stabilizes the system, with all states converging, albeit with increased oscillations.

Simulations under varying parameters reveal sensitivity to the friction coefficient b , where low damping leads to prolonged oscillations or instability. This highlights that the LMI-based controller guarantees local stability but is affected by parameter variations.

Overall, the simulation results provide preliminary validation of the proposed hybrid control framework, demonstrating that the model-based LMI controller stabilizes the system under nominal conditions while revealing sensitivity to parameter variations. These observations motivate the integration of data-driven control to improve adaptability under uncertain and varying conditions, forming a unified approach that combines model-based stability with data-driven flexibility. Future work will extend this framework by incorporating passivity constraints, along with experimental validation on a physical robot platform.

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2) *Results and Discussion:* To evaluate the accuracy of the linear approximation and controller performance, the re-